TEAM DAMI

Team Guide

DAMI

What we explain, how we explain it, and how we defend every number in front of the judges

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Jordan Students Pitch Competition — GEW Jordan

1 The project in one minute

DAMI is an AI platform and a medical-drone service that makes sure the right blood unit reaches the right hospital before it is needed. It forecasts demand for every blood type at every hospital, rebalances stock before units expire, calls the right donors, and in emergencies launches a cooled drone that carries blood from the blood bank to the hospital in minutes.

> **Elevator pitch (memorize): "Bleeding causes 30–40% of trauma deaths. DAMI uses AI to forecast and balance blood stock, call the right donors, and fly blood by cooled drone in minutes instead of half an hour."**

2 The problem (with verified facts)

- Hemorrhage causes 30–40% of trauma deaths, and 33–56% of those deaths happen before the patient reaches hospital (Kauvar, Lefering & Wade, Journal of Trauma, 2006).
- Most trauma deaths happen in the first hours after injury, so delivery time matters.
- Blood expires: platelets last only 5–7 days, red cells 35–42 days at 2–6 °C. One hospital runs short while another wastes expiring units.
- O negative is the type used first in severe bleeding before the patient's type is known, and it is scarce.
- Coordination today is mostly phone calls; general donation drives don't target the missing type.

> **Never say "3 out of 10 patients die from blood shortage". We have no source for that. Use the 30–40% figure and name the source.**

3 Why blood runs short

Cause	What it means
Seasonal donation	Donations drop in summer, Ramadan and exam periods
Short shelf life	Platelets 5–7 days; red cells 35–42 days
Rare types	O– is always needed and rarely available
Uneven distribution	Stock concentrates in the center; outskirts wait
Sudden emergencies	One major accident can use days of stock
Manual coordination	Phone calls, no live national view

These are general, well-known causes; we will confirm the Jordanian picture in interviews with blood banks.

4 The solution: 3 layers, 8 modules

Module	What it does	How (tech)

Smart inventory	Every unit tracked with type, expiry and location	RFID/barcode per unit + fridge sensors → live database
Demand forecasting	Predicts 14-day demand per type per hospital	Time-series models (Prophet / LightGBM) on history, weekdays, seasons
Smart balancing	Moves units close to expiry to a hospital that will use them	Optimization: minimize expected waste + shortage
Matching engine	Finds a compatible type and the nearest source	Blood compatibility rules + ranking by stock, distance and ETA
Fleet management	Plans safe drone routes; a licensed pilot approves	Route planning with restricted zones and weather; MAVLink
Cold chain	Box temperature every second, instant alarm, full log	IoT temperature loggers over 4G/5G
Donor engine	Invites eligible, nearby donors for the missing type	Eligibility (90+ days), distance ≤ 12 km, response-likelihood model
National dashboard	Live national view for the Ministry of Health	Aggregated, anonymized data

> **Key message: DAMI is not just a drone. The drone is only the emergency layer of a full blood-management system.**

5 Emergency request, step by step

Time	Step
0 s	Hospital requests 6 units O– in the app
5 s	AI checks live stock and compatibility
10 s	Best source chosen: drone or car (ETA comparison)
1 min	Licensed pilot approves; box packed at 2–6 °C
≈ 9 min	Flight with live GPS and temperature
✓	Delivery confirmed; donors called to restock

These times come from our simulation for a typical distance in Amman. Say “in our simulation”.

6 The drone and safety

- Multirotor drone for the pilot (payload 2–4 kg ≈ 4–6 units), fixed-wing VTOL later for longer distances.
- Cooling box with phase-change material keeps 2–6 °C; logger reports every second; any out-of-range shipment is flagged and not used.

- Safety: emergency parachute, automatic return on link loss, geofencing, routes away from crowds, liability insurance.
- Regulation: permits from the Civil Aviation Regulatory Commission (CARC) and security authorities. We start in simulation, then one approved pilot route.
- If flying is impossible (weather, fault), the system switches automatically to road transport from the nearest source.

7 Matching engine: the compatibility rules

- O− can give red cells to every type (universal donor).
- AB+ can receive from every type (universal recipient).
- Rh−negative patients receive only Rh−negative blood; Rh−positive patients can receive both.
- If the requested type runs out, the engine proposes the safest compatible alternative, and the final medical decision stays with the hospital.

8 Business Model Canvas, block by block

Block	Our answer	Why
Customer segments	Private hospitals first (71 of 121), then MoH hospitals, National Blood Bank; donors use it free	Private hospitals decide faster and pay for quality
Value propositions	Emergency blood in minutes; less waste; live national view; right donor at the right time	Saves lives and money
Channels	Direct sales to hospitals, Ministry partnership, mobile app + web dashboard	B2B health sales are relationship-based
Customer relationships	Annual service contract, 24/7 emergency center, monthly impact reports	Critical service needs trust
Revenue streams	Subscription JOD 6,000/yr; flight JOD 90; national contract JOD 40,000/yr; sponsorship JOD 10,000/yr	Recurring base + usage + government + CSR
Key activities	Forecasting & matching models, drone operations, cold-chain monitoring, donor call-ups	Core of the service
Key resources	AI platform, RFID per unit, drones & smart boxes, IoT sensors, team & licensed pilots	Hard to copy together
Key partners	MoH & National Blood Bank, hospitals, drone makers, CARC, Red Crescent & sponsors	Access, permits and credibility
Cost structure	Initial JOD 58,000; operating JOD 50K → 95K → 150K; ≈ JOD 40 direct per flight	Team is 60% of operating costs

9 The numbers — and how to defend each one

Judges said our first version was missing budget, costs and revenue detail. The new deck adds 6 slides: Market size, Competition, Revenue, Costs, Budget, Financial summary.

Item	Year 1	Year 2	Year 3	How we got it
Hospitals subscribed	3	10	20	Assumption: realistic sales pace for a new health startup
Subscription revenue	18,000	60,000	120,000	hospitals × JOD 6,000
Drone flights	0	400	1,500	Flights start in year 2 after permits
Flight revenue	0	36,000	135,000	flights × JOD 90
National dashboard	0	0	40,000	One government contract in year 3
Sponsorship	10,000	20,000	30,000	1 → 2 → 3 sponsors × JOD 10,000
Total revenue	28,000	116,000	325,000	Sum of the 4 streams
Operating result	-25,700	-6,200	85,250	Gross profit – operating costs

- Budget ask JOD 89,900 = initial investment JOD 58,000 (hardware 30,000 · software 23,000 · compliance 5,000) + JOD 31,900 to cover operating losses until month 24.
- Monthly cash flow turns positive from month 25.
- Break-even (all investment recovered): around month 37 in the base case, month 27 in the optimistic case (+40% volume).
- Gross margin in year 3 ≈ 72%. Each drone flight: price JOD 90, direct cost ≈ JOD 40 (battery wear, maintenance, pilot time, insurance) → ≈ 55% contribution margin.
- Market: Jordan has 121 hospitals (71 private, 33 MoH, 15 Royal Medical Services, 2 university — source: trade.gov Jordan Country Commercial Guide). TAM = 121 × 6,000 = JOD 726K/yr; SAM = 71 × 6,000 = JOD 426K/yr; SOM = 20 × 6,000 = JOD 120K/yr by year 3.
- Why JOD 6,000? That is JOD 500 a month per hospital. We will validate it in 10–15 interviews with hospital and blood-bank managers.

Always say: "These are planning assumptions; our next step is validating them with hospitals and suppliers." Honesty about assumptions scores better than fake certainty.

10 Who presents what (suggestion)

A suggested split for a 5-minute pitch (~1 minute each). Adjust to the competition's time limit; everyone should still be able to answer finance questions.

Who	Topic	Slides
Abdulrahman (Software Engineer)	Opening story, how the system works, live demo, closing	Slides 1, 6, 8–10, 13 (demo), 24–25

Marah (AI Engineer)	The problem and why blood runs short	Slides 2–4
Nadeen (AI Engineer)	The AI: forecasting, matching engine, smart donor	Slides 5, 7, 11–12
Darya (AI Engineer)	Proof, market size, competition, revenue model	Slides 14–18
Abdulmonem (AI Engineer)	Costs, budget, financial summary, plan and the ask	Slides 19–23

11 Live demo (2 minutes)

1. Open the DAMI app before entering (works offline).
2. Dashboard: point at the red alerts (critical shortages) and expiring units.
3. Emergency drone → Find best source → show drone vs car time.
4. Launch drone now → watch it fly; point at box temperature ≈ 4 °C.
5. Press Simulate cooling failure → the alarm appears and backup cooling activates.
6. Mass casualty → Run incident scenario → three hospitals served automatically.

12 Judge questions (with answers)

Q: Are drones allowed in Jordan?

A: They need CARC and security approvals. We start with software and simulation, then one approved pilot route; year-1 revenue doesn't depend on flying.

Q: How do you keep the blood safe?

A: Certified 2–6 °C box, temperature every second, instant alarm, full log with each shipment; out-of-range units are flagged and not used.

Q: What if a drone crashes?

A: Parachute, auto-return, geofencing, routes away from crowds, insurance, licensed pilot on every flight.

Q: Where does the data come from?

A: The prototype uses realistic synthetic data. In the pilot we connect partner hospitals under a formal agreement with encryption and consent.

Q: Why JOD 6,000 per hospital?

A: It's JOD 500 a month for less waste and faster emergency supply. We'll validate the price in 10–15 interviews.

Q: Why do you need JOD 89,900?

A: JOD 58,000 initial investment plus JOD 31,900 to run until we're cash-positive in month 25.

Q: When do you break even?

A: Around month 37 in the base case, month 27 in the optimistic case. Year 3 is already profitable.

Q: How big is the market?

A: 121 hospitals in Jordan → JOD 726K a year in subscriptions alone; we target 20 hospitals by year 3, plus flights and a national contract; later the region.

Q: Who are your competitors?

A: Manual coordination today; blood-bank software that records stock only; Zipline delivers but isn't in Jordan and doesn't manage stock.

Q: What if the Ministry doesn't sign?

A: Private hospitals are our first customers; the national contract is only year-3 upside.

Q: What's the biggest risk?

A: Drone licensing. That's why the software creates value first and drones come in year 2.

Q: How is AI used, really?

A: Forecasting demand per type, optimizing stock moves, ranking sources, and predicting which donors will respond.

Q: What have you built?

A: A working simulator: dashboard, forecasting, donors, emergency drone with temperature tracking, mass-casualty mode.

Q: How many units per flight?

A: 4–6 units in the pilot drone — enough for the first emergency need until the larger batch arrives by road.

Q: How do you measure success?

A: Units saved from expiry, minutes saved, % shipments in safe temperature, active donors.

Q: Why you?

A: A software engineer plus four AI engineers: we can build the platform and the models ourselves.

13 Say carefully / never claim

- Never invent statistics. Use only the sourced numbers in this guide.
- Say “per our preliminary research” when you say there is no similar system in Jordan.
- Say “in our simulation” for the 9-minute vs 34-minute times.
- Say “planning assumptions” for all financial figures.
- Medical decisions always stay with the hospital's medical staff.

14 Final checklist

☐ Everyone knows their slides and timing

☐ Demo rehearsed 3 times

☐ Laptop charged, app open, PDF backup on USB

☐ Each member can answer at least 3 judge questions

☐ Sources ready: Kauvar 2006, Nisingizwe 2022, trade.gov

★ Sources

1. Kauvar DS, Lefering R, Wade CE. Impact of hemorrhage on trauma outcome. J Trauma 2006;60(6 Suppl):S3–S11. pubmed.ncbi.nlm.nih.gov/16763478
2. Nisingizwe MP et al. Effect of unmanned aerial vehicle (drone) delivery on blood product delivery time and wastage in Rwanda. Lancet Glob Health 2022;10:e564–e569.

3. U.S. International Trade Administration. Jordan Country Commercial Guide — Healthcare. trade.gov/country-commercial-guides/jordan-healthcare